VIJAYSINGH PUWAR

Cybersecurity Engineer | IT Infrastructure, Networking and Secure Software
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SUMMARY

Cybersecurity engineer with IT infrastructure experience at the MTA and R.S. Infotech across identity and access, Cisco networking and Windows and Linux operations. Builds and runs production web and AI applications in Python, TypeScript, Java and Go. CompTIA Security+, CySA+ and Cisco CCNA.

EXPERIENCE

Emerging Talent Intern, Metropolitan Transportation Authority (MTA) New York, NY | Jun 2026 – Present

- Resolve incidents and service requests in ServiceNow on a 24/7 rail operation, isolating faults across application, network path, access policy and host, and escalating to senior staff or vendors (Dell, HP, Solari).
- Review Active Directory, DUO MFA and Zscaler ZPA access requests with the security team against policy.
- Troubleshoot Cisco Catalyst switches, VLANs, trunk ports and OSPF and BGP routing at sites across the railway; inventoried every network device and traced each one to its switch port.
- Built the SIR IT request intake system in Microsoft Forms, Power Automate and SharePoint, caught a defect silently writing incorrect values before release, and wrote the operator runbook.
- Wrote and deployed a Python NTP clock-correction service on a Rail Control Center workstation with dry-run mode, elevation check, rotating logs and distinct exit codes.

System Engineer, R.S. Infotech Vadodara, India | Feb 2023 – Aug 2024

- Ran 150+ Windows and Linux systems on premises and on AWS and Azure, including patching and scheduled changes.
- Managed Active Directory and Group Policy: MFA enrollment, privileged group reviews, onboarding and offboarding.
- Wrote Splunk SPL queries for failed authentication and suspicious activity, and tuned IDS/IPS and firewall rules to cut false positives.
- Triaged SonarQube, OWASP ZAP, Burp Suite and Nessus findings and added security checks to GitLab CI.
- Wrote Python, PowerShell and Bash scripts for compliance checks, log review and inventory.

PROJECTS

Recap Verse | AI video SaaS with paying subscribers recapverse.com

Python, FastAPI, Next.js, MongoDB, Redis, Google Cloud Run, Anthropic, OpenAI and Gemini APIs

- Built and run it alone: a multi-stage pipeline that combines LLM calls with deterministic stages, covered by 2,019 automated tests across five CI jobs.
- Provider-abstracted LLM layer across Anthropic, OpenAI and Gemini with schema-validated JSON output and quality-gated retries.
- Encrypt per-user API keys with AES-GCM, enforce resource ownership on every route, and gate CI on gitleaks secret scanning.
- Stripe and PayPal subscriptions through deduplicated webhooks, so a redelivered event is never applied twice.

Web Audio TV | Audiobook streaming platform, about 1,000 users webaudiotv.com

TypeScript, React, PostgreSQL with row level security (Supabase), Cloudflare Workers and R2

- Found and fixed three production security flaws: an email queue the public API key could call, a publicly downloadable storage bucket, and a leaderboard farming exploit traced with SQL over raw listening events.
- Serve audio only through a Cloudflare Worker that checks a short-lived signed ticket bound to caller, resource and expiry.
- Verify every database role's permissions against the real migrations with a 48-check access matrix; 1,059 automated tests.

BreachScope | Live threat-intelligence map GitHub

Go (chi, pgx), PostgreSQL, React, TypeScript, MapLibre GL, Vercel

- Ingests six public feeds (ThreatFox, URLhaus, Feodo Tracker, SANS DShield, CISA KEV, Have I Been Pwned) and records provenance and location precision on every record.
- Resumable SSE stream with Last-Event-ID; load tested at 20,000 requests and 200 concurrent with zero errors and zero data races.

TrustKart | Full-stack online store GitHub

Java 21, Spring Boot, PostgreSQL, Redis, React

- Checkout runs as one database transaction, so 100 concurrent shoppers cannot duplicate orders, oversell or deadlock; idempotency keys guarantee exactly-once orders.

Dual-ISP Network Assessment GitHub

Nmap, Wireshark, PowerShell, Bash, Python

- Traced conflicting device inventories to two ISP routers on the same 192.168.0.0/24 subnet (one gateway IP answering from two MACs), moved one LAN to 192.168.20.0/24 and confirmed one gateway MAC, one DHCP server and no duplicate addresses.

EDUCATION AND CERTIFICATIONS

M.S. Cybersecurity, Pace University, Seidenberg School of CSIS | GPA 3.92 Dec 2026 (expected)

Coursework: Ethical Hacking, Network Security and Defense, Automating Information Security with Python and PowerShell, Cyber Intelligence Analysis, Algorithms, Object-Oriented Programming

B.E. Mechanical Engineering, G H Patel College of Engineering and Technology Aug 2023

Certifications: CompTIA Security+, CompTIA CySA+, Cisco CCNA, Google AI Essentials

SKILLS AND ACTIVITIES

Security and Identity: Splunk (SPL), IDS/IPS, firewalls, Burp Suite, OWASP ZAP, Nessus, Active Directory, Group Policy, DUO MFA, Zscaler ZPA

Networking and Systems: TCP/IP, DNS, DHCP, VLANs, OSPF, BGP, Cisco Catalyst, Linux, Windows Server, ServiceNow

Cloud and DevOps: AWS, Azure, Google Cloud Run, Cloudflare, Terraform, Docker, GitHub Actions, GitLab CI

Languages and Frameworks: Python, TypeScript, Java, Go, C#, Rust, SQL, PowerShell, Bash, FastAPI, Spring Boot, React, Next.js

AI: Anthropic, OpenAI and Gemini APIs, structured-output prompting, schema validation, evaluation harnesses

Databases: PostgreSQL, MongoDB, Redis

Activities: Pace University Cybersecurity Club; STEM outreach volunteer mentoring undergraduate and high school students in cybersecurity; TryHackMe top 8%, 217 active days in 2025